

LinkForge Manual & Runbook

Master Guide: GitHub Synchronization, Antigravity Development, Fresh Windows Re-Installation & Remote Deployment

Local Project Path:	X:\.config\docker\linkforge
GitHub Repository:	https://github.com/siegewang/Linkforge.git
Active Port / URL:	http://localhost:5006 (or http://<IP>:5006)
Database & Archives:	X:\.config\docker\linkforge\data\linkforge.db

1. Fresh Windows Setup & Clean Re-Installation

If you ever reinstall Windows from scratch and need to get LinkForge running immediately, follow these four simple steps:

- **Step 1: Install Prerequisites:**

- **Docker Desktop:** Download and install Docker Desktop for Windows (enable WSL 2 backend during setup).
- **Git for Windows:** Download and install Git from git-scm.com.
- **Google Antigravity / Python 3.12:** Install Google Antigravity coding environment.

- **Step 2: Clone the Project from GitHub:**

Open PowerShell as Administrator and run the following commands to create the folder and clone your repository:

```
mkdir -p X:\.config\docker
cd X:\.config\docker
git clone https://github.com/siegewang/Linkforge.git linkforge
cd linkforge
```

- **Step 3: Launch the Container:**

From inside X:\.config\docker\linkforge, start the Docker container:

```
docker compose up -d --build
```

- **Step 4: Restore Data (Optional):**

Open your browser to <http://localhost:5006/admin/data>, click **Restore from Backup / CSV**, and select your latest `linkforge_auto_backup_*.zip` file. All bookmarks, offline articles, and video categories will be populated automatically.

2. How to Make Changes & Develop with Antigravity

When you want to add features, modify styling, adjust AI ingestion, or fix anything in LinkForge with Antigravity:

- **Opening the Project in Antigravity:**

1. Launch Google Antigravity.
2. Open the directory workspace: X:\.config\docker\linkforge.

3. AntigraVity automatically detects all blueprints, services, templates, and extension folders.

Example Prompts to Give AntigraVity:

- *'Add a new tag filter in the Neural Links sidebar that sorts tags by frequency.'*
- *'Update the YouTube transcript extractor to support additional subtitle languages.'*
- *'Change the theme accent colors in base.html and ensure mobile responsive padding is clean.'*
- *'Package the browser extensions and update both Chrome and Firefox ZIP bundles.'*

Codebase Structure for Reference:

Directory / File	Description & Responsibility
blueprints/	Flask route controllers (dashboard, bookmarks, videos, notes, admin, api)
services/	Core logic (backup.py, updater.py, nuke.py, db.py, extractors.py)
templates/	Jinja2 HTML frontend views (base.html, settings.html, admin_*.html)
data/	SQLite database (linkforge.db), backups/, and offline archives/ (Ignored by Git)
docker-compose.yml	Container definition, port mapping (5006:5000), and volume bindings

3. How to Update Your GitHub Repository

Whenever you or Antigravity complete code changes, push them to GitHub with this standard 3-command workflow:

The 3-Step Git Publish Workflow:

```
# 1. Navigate to the project folder
cd X:\.config\docker\linkforge

# 2. Stage all modified files and commit with a description
git add .
git commit -m "Describe what you changed here (e.g. Add dark mode toggle)"

# 3. Push to your GitHub repository
git push origin main
```

100% Privacy Guarantee:

The .gitignore configuration strictly excludes your ./data/ folder, SQLite databases (*.db), cached article archives, and private secrets. Pushing to GitHub will **NEVER** upload your bookmarks, personal notes, or login credentials.

4. Remote Deployment (e.g., Dad's PC) & Live Updates

How remote instances receive updates seamlessly without ever touching the terminal:

Initial One-Time Setup on Dad's Computer:

1. Clone the repository: `git clone https://github.com/siegewang/Linkforge.git`
2. Launch Docker: `docker compose up -d --build`
3. Done! The auto-updater service is now active inside his container.

How Future Updates Work (Zero-Terminal / 100% In-Browser):

1. You push new code to GitHub from your PC (`git push origin main`).
2. Within seconds, Dad's LinkForge will detect the new GitHub commit.
3. A glowing green/cyan badge **■ UPDATE AVAILABLE** appears in his left sidebar.
4. In **Settings**, a single button appears: **Update & Restart Now**.
5. Dad clicks the button — LinkForge pulls the update, restarts in 2 seconds, and reloads with zero data loss!

Cleaning Test Data (The Nuke Data Tab):

To clean any restored test bookmarks on Dad's computer while leaving all his AI configs intact:

1. Open **Admin Panel** → **Nuke Data** (/admin/nuke).
2. Click '**Preset: Clean Restored Data (Keep AI Settings)**'.
3. Type **NUKE** in the confirmation box and click **Confirm Nuke**.
4. All personal links, videos, and articles are wiped clean, while the Smart Master Switch, YouTube fetchers, and AI routing remain 100% saved.

5. Troubleshooting & Maintenance Commands Cheat Sheet

Action / Problem	Command to Run in PowerShell
Restart LinkForge Container	<code>docker compose restart</code>
Rebuild from scratch (forced)	<code>docker compose up -d --build --force-recreate</code>
View live application logs	<code>docker logs -f linkforge</code>
Discard local unpushed changes	<code>git reset --hard origin/main</code>
Repackage browser extensions	<code>python package_extensions.py</code>